Supplementary Material for "Certified Federated Unlearning for IoT: Ledger-Anchored Revocation with Anytime-Valid Forgetting Certificates"

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I. Setup, cost model and attack definitions

All experiments are produced by the scripts in the repository (`experiments/unlearning_experiments.py`, `unlearning_real.py`, `unlearning_new.py`, `unlearning_review.py`); the tables below are written from the result files by `paper/make_supp_tables.py`.

Training. Binary logistic loss, full-batch local gradient, update clipping at norm 1, learning rate 1, $T = 60$ rounds, participation $p = 0.6$ (independent per client and round), reputation rate 0.1 and penalty 0.2, Pedersen commitments with a 512-bit group, spectral split gated by a separation margin of 0.25, no differential-privacy noise unless stated. Synthetic fleet: 60 clients, $d = 1024$, 12 real records and 6 canary pairs per client, Dirichlet(α=1) over ten latent classes, 8% label noise. UCI HAR: 30 subjects, 40 records each. N-BaIoT: nine devices, five sites each, 40 records each, features $\mathrm{sign}(x)\log(1+|x|)$ standardised. The MLP has 32 tanh hidden units. Test records are held out per client. Audits use $\epsilon = 0.2$, $\alpha = 0.05$ and, unless stated, a reveal order drawn from a fixed seed.

Cost model. A client round with n records costs $n \cdot f$ FLOPs ($f = 4d$ for the linear model, $6P$ for an MLP with P parameters), uploads $4P + 64$ bytes (float32 update and a 512-bit commitment) and downloads $4P$ bytes. Energy is $e_{\text{flop}} \cdot \text{FLOPs} + e_{\text{bit}} \cdot 8 \cdot \text{bytes}$ with $e_{\text{flop}} = 1$ nJ and $e_{\text{bit}} = 100$ nJ; these are assumed constants, not measurements, and Table S10 varies each over a hundred-fold range. The canary reveal ($2m(d{+}1) \cdot 4$ bytes per revoked client) is added to every policy. Dropped clients are not charged. Server computation and storage are reported separately (Table S9).

Early-stopped retraining. From the initial model on the retained clients, blocks of 5 rounds are run until validation accuracy (150 held-out records) is within 0.5 points of the model before unlearning, up to 60 rounds.

Dynamics. Dropout: each selected client fails with probability 0.3 before uploading (at least three survive). Asynchrony: each client computes its gradient at the model of a round drawn uniformly from the last four. Staggered enrolment: client i of N may take part only from round $\lfloor 40(N-1-i)/(N-1) \rfloor$, and the eight last clients from round 0; the wave consists of the first (latest-enrolled) clients.

Attacks. *Crossover canaries:* each coordinate of a canary is taken from one of two random records of the client, label random. *Filter:* an isolation forest (100 trees) is fitted to all training records, and the operator removes as many highest-scoring records as there are canaries before training the unlearned model. *Selective reveal:* the revoking clients open all pairs with a non-positive loss difference and an equal number of positive ones; the accept-partial verifier ignores unopened pairs, the strict verifier scores them as extreme. *Ledger:* a record in the middle of the chain is edited and every later hash is recomputed; three witnesses store the heads published at the anchor interval.

II. Additional figures

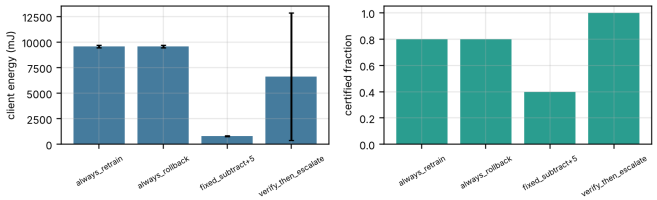


Fig. S1. Synthetic fleet, wave of 20 clients (10 seeds): mean client energy (bars, s.d. as error bars) and share of waves certified for the verify-then-escalate ladder, always retraining, always rollback and a fixed subtract+5 policy.

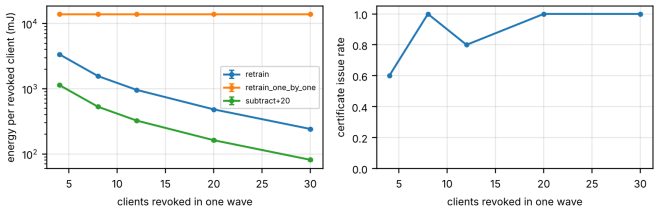


Fig. S2. Synthetic fleet: energy per revoked client versus wave size (left; "one by one" is measured on three requests and extrapolated) and issue rate of the certificate for the retrained model (right, 5 seeds).

III. Additional tables

IV. Reproduction

```
pip install -r requirements.txt
python experiments/unlearning_experiments.py    # s
python experiments/unlearning_real.py --dataset ha
python experiments/prepare_nbaiot.py            # o
python experiments/unlearning_real.py --dataset nb
```

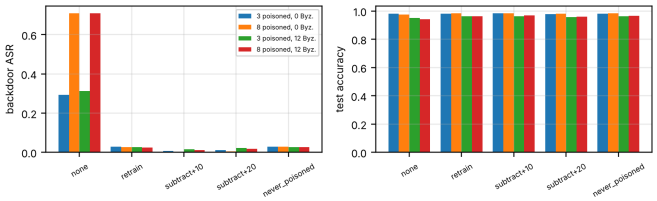


Fig. S3. Synthetic fleet, backdoor success rate and test accuracy before and after unlearning, with 3 or 8 poisoned clients in the wave and with 0 or 12 Byzantine sign-flipping clients (5 seeds).

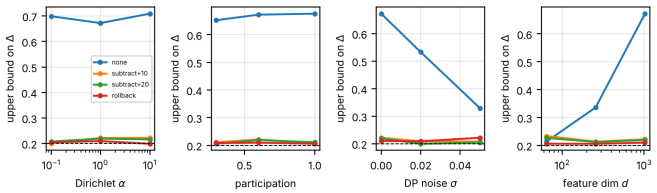


Fig. S4. Synthetic fleet: upper bound on the residual advantage against the Dirichlet parameter, participation rate, DP noise and feature dimension for the un-unlearned model and three operators (5 seeds). Dashed: $\epsilon = 0.2$.

```
python experiments/unlearning_new.py
python experiments/unlearning_review.py
python experiments/fig_new.py
python paper/make_supp_tables.py
python tests/test_unlearning.py
```

TABLE S1
SYNTHETIC FLEET, ONE WAVE OF 20 OF 60 CLIENTS, 10 SEEDS. I/U/R: CERTIFICATE ISSUED, UNDETERMINED, REVOKED. Δ_{up}: MEAN UPPER BOUND ON THE RESIDUAL ADVANTAGE.

Operator	I/U/R	Acc.	Δ_{up}	Rounds	MB	Energy (J)
none	0/0/10	0.981	0.667	0	0.00	0.00
subtract	4/1/5	0.620	0.243	0	0.00	0.00
replay	4/1/5	0.621	0.252	0	0.00	0.00
subtract+5	4/1/5	0.983	0.249	5	1.00	0.81
subtract+10	4/3/3	0.982	0.226	10	2.03	1.64
subtract+20	6/3/1	0.981	0.212	20	4.00	3.24
rollback	8/2/0	0.983	0.212	60	11.84	9.58
retrain	8/2/0	0.983	0.212	60	11.84	9.58

TABLE S2
CALIBRATION: THE UN-UNLEARNED MODEL (SHOULD BE FLAGGED) AND THE EXACTLY RETRAINED MODEL (SHOULD NOT).

Data	Model	Model state	Runs	I/U/R
synthetic	linear	none	40	0/0/40
synthetic	linear	retrain	40	40/0/0
HAR	linear	none	60	0/0/60
HAR	linear	retrain	60	49/11/0
HAR	mlp	none	60	0/0/60
HAR	mlp	retrain	60	51/7/2
N-BaIoT	linear	none	60	42/9/9
N-BaIoT	linear	retrain	60	55/3/2
N-BaIoT	mlp	none	60	14/11/35
N-BaIoT	mlp	retrain	60	54/6/0

ICR_twins · revision-round experiments

TABLE S3
REAL DATA, A WAVE OF ONE THIRD OF THE CLIENTS, 5 SEEDS.

Data	Model	Operator	I/U/R	Acc.	Rounds	Energy (J)
HAR	linear	none	0/0/5	0.999	0	0.00
HAR	linear	subtract	2/2/1	0.991	0	0.00
HAR	linear	replay	2/1/2	0.991	0	0.00
HAR	linear	subtract+5	2/2/1	0.998	5	0.22
HAR	linear	subtract+10	2/2/1	0.998	10	0.46
HAR	linear	subtract+20	3/1/1	0.998	20	0.91
HAR	linear	retrain	3/2/0	0.998	60	2.69
HAR	mlp	none	0/0/5	0.999	0	0.00
HAR	mlp	subtract	3/2/0	0.995	0	0.00
HAR	mlp	replay	3/2/0	0.995	0	0.00
HAR	mlp	subtract+5	4/1/0	0.998	5	7.21
HAR	mlp	subtract+10	4/1/0	0.998	10	14.90
HAR	mlp	subtract+20	4/1/0	0.998	20	29.24
HAR	mlp	retrain	4/1/0	0.999	60	86.48
N-BaIoT	linear	none	3/1/1	0.996	0	0.00
N-BaIoT	linear	subtract	4/1/0	0.871	0	0.00
N-BaIoT	linear	replay	4/1/0	0.942	0	0.00
N-BaIoT	linear	subtract+5	4/1/0	0.960	5	0.08
N-BaIoT	linear	subtract+10	4/1/0	0.995	10	0.15
N-BaIoT	linear	subtract+20	4/1/0	0.996	20	0.30
N-BaIoT	linear	retrain	4/1/0	0.995	60	0.88
N-BaIoT	mlp	none	0/0/5	0.997	0	0.00
N-BaIoT	mlp	subtract	5/0/0	0.962	0	0.00
N-BaIoT	mlp	replay	4/1/0	0.939	0	0.00
N-BaIoT	mlp	subtract+5	5/0/0	0.996	5	2.38
N-BaIoT	mlp	subtract+10	5/0/0	0.997	10	4.73
N-BaIoT	mlp	subtract+20	5/0/0	0.997	20	9.35
N-BaIoT	mlp	retrain	5/0/0	0.997	60	27.36

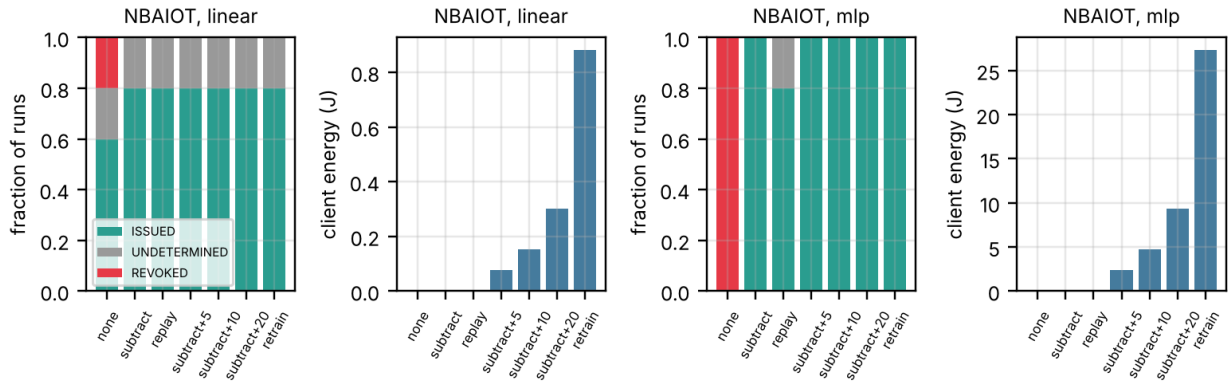


Fig. S5. N-BaIoT (45 clients, wave of 15, 5 seeds): certificate outcomes and client energy per operator for the linear model and the MLP.

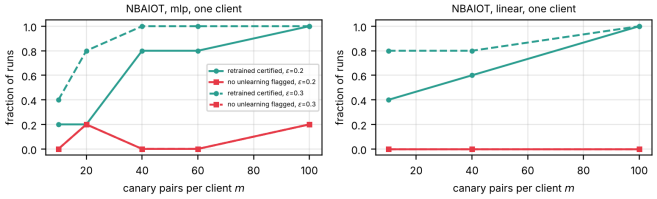


Fig. S6. N-BaIoT, one revoked client: share of runs in which the retrained model is certified and the un-unlearned model is flagged, against canary pairs per client. The audit has little power at these model sizes.

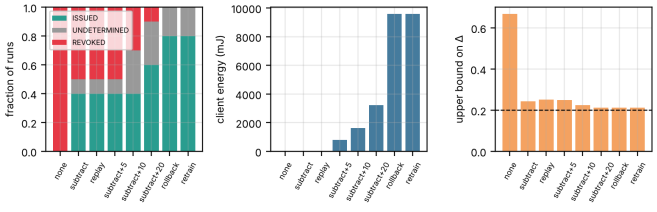


Fig. S7. Synthetic fleet (10 seeds): certificate outcomes, client energy and upper bound on the residual advantage per operator.

TABLE S4
RECOVERY POLICIES (5 SEEDS). ENERGY INCLUDES THE CANARY REVEAL. U COUNTS UNDETERMINED AND UNPOWERED. LAST COLUMNS: WORST RUN, AND WORST SINGLE DEVICE.

Data	Model	Policy	I/U/R	Acc.	Rounds	Mean J	Max J	Dev. J
Synthetic	linear	retrain (60)	5/0/0	0.986	60.0	10.42	10.66	0.297
Synthetic	linear	retrain early	5/0/0	1.000	5.0	1.59	1.66	0.034
Synthetic	linear	ladder	5/0/0	0.985	41.0	7.46	14.60	–
Synthetic	linear	ICR	5/0/0	0.982	17.0	3.53	5.62	0.102
Synthetic	linear	ICR q=0.5	4/1/0	0.982	16.0	2.08	5.59	0.088
Synthetic	linear	ICR q=0.25	4/1/0	0.988	26.0	1.82	3.15	0.127
HAR	linear	retrain (60)	3/2/0	0.999	60.0	2.91	2.96	0.156
HAR	linear	retrain early	4/0/1	0.998	16.0	0.94	2.96	0.046
HAR	linear	ladder	3/2/0	0.996	39.0	1.96	4.03	–
HAR	linear	ICR	4/1/0	0.996	24.0	1.29	2.85	0.119
HAR	linear	ICR q=0.5	4/1/0	0.996	18.0	0.61	1.51	0.084
HAR	linear	ICR q=0.25	3/2/0	0.996	27.0	0.55	0.97	0.141
HAR	mlp	retrain (60)	4/1/0	0.999	60.0	86.70	88.34	5.015
HAR	mlp	retrain early	4/1/0	0.999	5.0	7.45	8.29	0.579
HAR	mlp	ladder	4/1/0	0.997	18.0	25.77	119.80	–
HAR	mlp	ICR	4/1/0	0.997	13.0	18.76	84.72	3.074
HAR	mlp	ICR q=0.5	4/1/0	0.997	13.0	9.23	41.68	2.773
HAR	mlp	ICR q=0.25	4/1/0	0.997	13.0	5.30	23.84	3.315
N-BaIoT	linear	retrain (60)	4/1/0	0.995	60.0	0.95	0.96	0.035
N-BaIoT	linear	retrain early	5/0/0	0.991	17.0	0.31	0.37	0.012
N-BaIoT	linear	ladder	1/4/0	0.980	24.0	0.42	1.32	–
N-BaIoT	linear	ICR	1/4/0	0.980	16.0	0.31	0.97	0.013
N-BaIoT	linear	ICR q=0.5	1/4/0	0.981	15.0	0.18	0.52	0.012
N-BaIoT	linear	ICR q=0.25	1/4/0	0.980	10.0	0.11	0.20	0.008
N-BaIoT	mlp	retrain (60)	5/0/0	0.997	60.0	27.43	27.81	1.092
N-BaIoT	mlp	retrain early	5/0/0	0.992	8.0	3.63	4.56	0.182
N-BaIoT	mlp	ladder	5/0/0	0.982	2.0	0.95	2.44	–
N-BaIoT	mlp	ICR	5/0/0	0.982	2.0	0.95	2.44	0.126
N-BaIoT	mlp	ICR q=0.5	5/0/0	0.982	2.0	0.49	1.20	0.126
N-BaIoT	mlp	ICR q=0.25	5/0/0	0.982	2.0	0.30	0.67	0.114

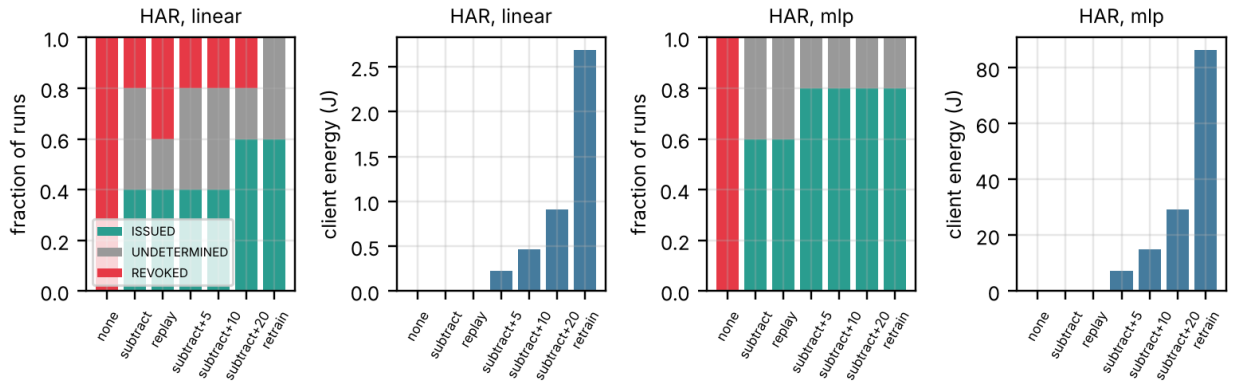


Fig. S8. UCI HAR: certificate outcomes and client energy per operator for the linear model and the MLP (5 seeds).

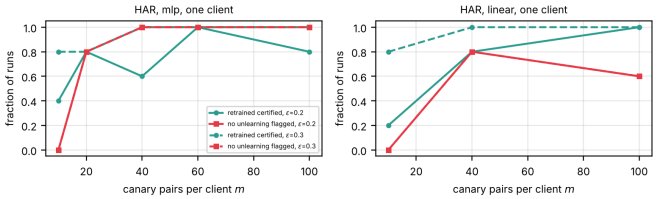


Fig. S9. One revoked client on UCI HAR: share of runs in which the retrained model is certified and the un-unlearned model is flagged, against canary pairs per client (5 seeds).

TABLE S5
CANARY DESIGN GRID: SHARE OF RUNS (5 SEEDS) IN WHICH THE AUDIT FLAGS THE UN-UNLEARNED MODEL.

Data	Model	m	Scale	Flagged	Δ_{up}	Acc.
N-BaIoT	linear	6	1	0.2	0.25	0.996
N-BaIoT	linear	12	1	0.6	0.22	0.996
N-BaIoT	linear	6	2	0.2	0.26	0.996
N-BaIoT	linear	12	2	0.6	0.21	0.967
N-BaIoT	linear	6	4	0.4	0.24	0.995
N-BaIoT	linear	12	4	0.6	0.21	0.932
N-BaIoT	linear	6	8	0.2	0.23	0.995
N-BaIoT	linear	12	8	0.8	0.22	0.967
N-BaIoT	mlp	6	1	1.0	0.36	0.997
N-BaIoT	mlp	12	1	1.0	0.35	0.997
N-BaIoT	mlp	6	2	1.0	0.50	0.997
N-BaIoT	mlp	12	2	1.0	0.52	0.997
N-BaIoT	mlp	6	4	1.0	0.57	0.997
N-BaIoT	mlp	12	4	1.0	0.59	0.997
N-BaIoT	mlp	6	8	1.0	0.46	0.997
N-BaIoT	mlp	12	8	1.0	0.56	0.997
HAR	linear	6	1	1.0	0.83	0.999
HAR	linear	12	1	1.0	0.65	0.999
HAR	linear	6	2	1.0	0.81	0.999
HAR	linear	12	2	1.0	0.61	0.999
HAR	linear	6	4	1.0	0.72	0.999
HAR	linear	12	4	1.0	0.61	0.999
HAR	linear	6	8	1.0	0.60	0.999
HAR	linear	12	8	1.0	0.56	0.999

TABLE S6
WAVE SCHEDULER SIMULATION, ONE YEAR, 20 RUNS, DEADLINE 30 DAYS, POWER TARGET 120 PAIRS.

Req./day	Policy	Requests	Waves	Mean lat.	Max lat.	J/request	Underpowered
0.2	immediate	73	66.2	0.5	1.0	2.48	1.00
0.2	batched	73	11.1	17.2	29.9	0.42	1.00
1	immediate	360	230.8	0.5	1.0	1.76	1.00
1	batched	360	18.1	10.3	29.6	0.14	0.07
4	immediate	1458	358.7	0.5	1.0	0.67	1.00
4	batched	1458	66.8	2.8	27.5	0.13	0.01

TABLE S7
BASELINES ON THE MLP (5 SEEDS): MARKER SUCCESS RATE, LOSS-THRESHOLD MEMBERSHIP AUC AND OUR CERTIFICATE OUTCOME.

Data	Model state	Marker ASR	MIA AUC	Canary I/U/R
HAR	none	0.034	0.438	0/0/5
HAR	retrain	0.034	0.438	4/1/0
HAR	subtract+20	0.030	0.438	4/1/0
N-BaIoT	none	0.395	0.426	0/0/5
N-BaIoT	retrain	0.432	0.416	5/0/0
N-BaIoT	subtract+20	0.442	0.414	5/0/0

TABLE S8
PROVENANCE LEDGER ON THE SYNTHETIC FLEET (3 SEEDS, 200 TAMPER TRIALS EACH).

Quantity	Value
records (60 rounds)	61
ledger size (kB)	101
bytes per round	1652
ledger / update traffic	0.011
chain verification (ms)	1.6
Merkle proof check (μs)	4.8
proof length (hashes)	5.7
tampering detected	1.00
checkpoints (kB)	417
stored updates (kB)	15154

TABLE S9
ENERGY ACCOUNTING (5 SEEDS): TRAFFIC, COMPUTATION, CANARY REVEAL TRAFFIC, SERVER STORAGE OF STORED UPDATES, AND ENTANGLEMENT-SCORE COMPUTATION.

Data	Model	Policy	MB	GFLOP	Reveal MB	Stored MB	Entangle GFLOP
Synthetic	linear	retrain (60)	11.91	0.11	0.98	17.7	0.00
Synthetic	linear	retrain early	0.99	0.01	0.98	17.7	0.00
Synthetic	linear	ladder	8.25	0.07	0.98	17.7	0.00
Synthetic	linear	ICR	3.38	0.03	0.98	17.7	0.07
Synthetic	linear	ICR q=0.5	1.60	0.01	0.98	17.7	0.07
Synthetic	linear	ICR q=0.25	1.27	0.01	0.98	17.7	0.07
HAR	linear	retrain (60)	3.27	0.07	0.27	4.8	0.00
HAR	linear	retrain early	0.88	0.02	0.27	4.8	0.00
HAR	linear	ladder	2.12	0.05	0.27	4.8	0.00
HAR	linear	ICR	1.30	0.03	0.27	4.8	0.02
HAR	linear	ICR q=0.5	0.48	0.01	0.27	4.8	0.02
HAR	linear	ICR q=0.25	0.41	0.01	0.27	4.8	0.02
HAR	mlp	retrain (60)	103.63	3.57	0.27	155.5	0.00
HAR	mlp	retrain early	8.67	0.30	0.27	155.5	0.00
HAR	mlp	ladder	30.62	1.06	0.27	155.5	0.00
HAR	mlp	ICR	22.22	0.77	0.27	155.5	0.58
HAR	mlp	ICR q=0.5	10.81	0.37	0.27	155.5	0.58
HAR	mlp	ICR q=0.25	6.10	0.21	0.27	155.5	0.58
N-BaIoT	linear	retrain (60)	1.07	0.02	0.08	1.5	0.00
N-BaIoT	linear	retrain early	0.30	0.01	0.08	1.5	0.00
N-BaIoT	linear	ladder	0.43	0.01	0.08	1.5	0.00
N-BaIoT	linear	ICR	0.29	0.01	0.08	1.5	0.01
N-BaIoT	linear	ICR q=0.5	0.14	0.00	0.08	1.5	0.01
N-BaIoT	linear	ICR q=0.25	0.05	0.00	0.08	1.5	0.01
N-BaIoT	mlp	retrain (60)	32.79	1.13	0.08	48.9	0.00
N-BaIoT	mlp	retrain early	4.28	0.15	0.08	48.9	0.00
N-BaIoT	mlp	ladder	1.06	0.04	0.08	48.9	0.00
N-BaIoT	mlp	ICR	1.06	0.04	0.08	48.9	0.18
N-BaIoT	mlp	ICR q=0.5	0.51	0.02	0.08	48.9	0.18
N-BaIoT	mlp	ICR q=0.25	0.28	0.01	0.08	48.9	0.18

TABLE S10
SENSITIVITY OF THE ENERGY SAVINGS (PERCENT) TO $e_{\mathrm{flop}} \in [0.1, 10]$ NJ AND $e_{\mathrm{bit}} \in [10, 1000]$ NJ/BIT. NEGATIVE: ICR COSTS MORE.

Data	Model	ICR vs retrain (60)	ICR vs early	ICR q=0.25 vs early
Synthetic	linear	66 to 69	-165 to -122	-20 to -15
HAR	linear	56 to 59	-44 to -36	41 to 50
HAR	mlp	78 to 79	-155 to -152	29 to 29
N-BaIoT	linear	68 to 71	2 to 3	66 to 78
N-BaIoT	mlp	97 to 97	74 to 75	92 to 93

TABLE S11
INDEPENDENT FORGETTING CRITERIA (5 SEEDS): PARAMETER DISTANCE TO A RETRAINED MODEL IN UNITS OF THE RETRAINING NOISE FLOOR, PREDICTION DISAGREEMENT WITH IT, LOSS ON THE REVOKED CLIENTS' REAL RECORDS RELATIVE TO IT, TEST ACCURACY.

Data	Model	Method	I/U/R	Gap	Disagree	Loss ratio	Acc.
Synthetic	linear	none	0/0/5	8.13	0.023	0.18	0.983
Synthetic	linear	subtract	2/1/2	6.80	0.370	8.00	0.620
Synthetic	linear	subtract+5	2/1/2	4.76	0.014	1.94	0.984
Synthetic	linear	subtract+10	2/3/0	4.15	0.011	1.72	0.984
Synthetic	linear	subtract+20	3/2/0	3.69	0.010	1.53	0.982
Synthetic	linear	icr_q1.0	5/0/0	4.26	0.010	1.77	0.982
Synthetic	linear	icr_q0.25	4/1/0	5.60	0.015	1.94	0.988
Synthetic	linear	retrain_early	5/0/0	9.79	0.014	0.64	1.000
Synthetic	linear	retrain_seed3	5/0/0	1.04	0.004	1.00	0.985
HAR	linear	none	0/0/5	8.12	0.000	0.05	0.999
HAR	linear	subtract	2/2/1	3.99	0.007	3.37	0.992
HAR	linear	subtract+5	2/2/1	2.98	0.001	1.21	0.998
HAR	linear	subtract+10	2/2/1	2.62	0.001	1.11	0.998
HAR	linear	subtract+20	3/1/1	2.29	0.000	1.03	0.999
HAR	linear	icr_q1.0	4/0/1	3.14	0.003	1.42	0.996
HAR	linear	icr_q0.25	3/1/1	3.93	0.003	1.66	0.996
HAR	linear	retrain_early	4/0/1	7.33	0.001	1.00	0.998
HAR	linear	retrain_seed3	3/2/0	0.87	0.000	0.98	0.999
HAR	mlp	none	0/0/5	6.61	0.000	0.05	0.999
HAR	mlp	subtract	3/2/0	3.85	0.003	1.88	0.996
HAR	mlp	subtract+5	4/1/0	3.19	0.001	1.17	0.998
HAR	mlp	subtract+10	4/1/0	2.85	0.001	1.08	0.998
HAR	mlp	subtract+20	4/1/0	2.48	0.001	1.03	0.998
HAR	mlp	icr_q1.0	4/1/0	3.44	0.002	1.27	0.997
HAR	mlp	icr_q0.25	4/0/1	3.66	0.002	1.47	0.997
HAR	mlp	retrain_early	4/1/0	9.17	0.000	1.17	0.999
HAR	mlp	retrain_seed3	4/1/0	0.93	0.000	0.99	0.999
N-BaIoT	linear	none	3/1/1	2.78	0.001	0.92	0.995
N-BaIoT	linear	subtract	4/1/0	2.88	0.136	6.51	0.865
N-BaIoT	linear	subtract+5	4/1/0	1.85	0.038	2.95	0.958
N-BaIoT	linear	subtract+10	4/1/0	1.41	0.002	1.17	0.995
N-BaIoT	linear	subtract+20	4/1/0	1.18	0.001	1.08	0.995
N-BaIoT	linear	icr_q1.0	4/1/0	2.04	0.017	1.58	0.980
N-BaIoT	linear	icr_q0.25	5/0/0	2.45	0.017	1.63	0.980
N-BaIoT	linear	retrain_early	5/0/0	3.12	0.005	1.34	0.991
N-BaIoT	linear	retrain_seed3	4/1/0	1.15	0.000	0.98	0.995
N-BaIoT	mlp	none	0/0/5	2.03	0.000	0.81	0.997
N-BaIoT	mlp	subtract	5/0/0	1.93	0.035	6.54	0.962
N-BaIoT	mlp	subtract+5	5/0/0	1.54	0.001	1.64	0.997
N-BaIoT	mlp	subtract+10	5/0/0	1.36	0.000	1.33	0.997
N-BaIoT	mlp	subtract+20	5/0/0	1.21	0.000	1.13	0.997
N-BaIoT	mlp	icr_q1.0	5/0/0	1.76	0.015	4.04	0.982
N-BaIoT	mlp	icr_q0.25	5/0/0	1.80	0.015	4.08	0.982
N-BaIoT	mlp	retrain_early	5/0/0	3.42	0.005	4.53	0.992
N-BaIoT	mlp	retrain_seed3	5/0/0	0.89	0.000	1.01	0.997

TABLE S12
CALIBRATION WITH AN UNPREDICTABLE (OS ENTROPY) REVEAL ORDER, $T = 30$.

Data	Model	State	Runs	I/U/R
Synthetic	linear	none	60	0/0/60
Synthetic	linear	retrain	60	60/0/0
HAR	mlp	none	60	0/0/60
HAR	mlp	retrain	60	51/9/0
N-BaIoT	mlp	none	60	14/10/36
N-BaIoT	mlp	retrain	60	53/6/1

TABLE S13
DROPOUT, ASYNCHRONY AND STAGGERED ENROLMENT (5 SEEDS): CERTIFICATE OUTCOMES, ICR ROUNDS, ICR ENERGY AS PERCENT OF RETRAINING, ROLLBACK CHECKPOINT AND ROLLBACK ROUNDS.

Data	Model	Variant	Retrain I/U/R	ICR I/U/R	ICR rounds	ICR %	Ckpt	Rollback rounds
Synthetic	linear	base	5/0/0	5/0/0	17	28	0	60
Synthetic	linear	dropout0.3	5/0/0	5/0/0	14	23	0	60
Synthetic	linear	stale3	5/0/0	5/0/0	7	12	0	60
Synthetic	linear	dropout+stale	5/0/0	5/0/0	6	10	0	60
Synthetic	linear	staggered	5/0/0	5/0/0	4	8	26	34
HAR	linear	base	3/2/0	4/1/0	24	40	0	60
HAR	linear	dropout0.3	3/2/0	4/1/0	17	29	0	60
HAR	linear	stale3	3/2/0	4/1/0	22	36	0	60
HAR	linear	dropout+stale	3/2/0	4/1/0	21	35	0	60
HAR	linear	staggered	4/1/0	4/1/0	12	24	26	34
HAR	mlp	base	4/1/0	4/1/0	13	21	0	60
HAR	mlp	dropout0.3	4/1/0	4/1/0	13	22	0	60
HAR	mlp	stale3	3/2/0	5/0/0	9	15	0	60
HAR	mlp	dropout+stale	3/2/0	4/1/0	12	20	0	60
HAR	mlp	staggered	4/1/0	4/1/0	12	24	26	34

TABLE S14
COHORT-SHARDED TRAINING (4 COHORTS, 5 SEEDS): WAVE INSIDE ONE COHORT OR SPREAD.

Data	Model	Wave	Clients	Cohorts	Sharded J	Mono J	Acc. sh.	Acc. mono	I/U/R
Synthetic	linear	spread	20	4	9.66	9.64	1.000	0.986	5/0/0
Synthetic	linear	one_shard	5	1	2.35	13.28	1.000	0.983	2/3/0
HAR	linear	spread	10	4	3.07	2.69	0.999	0.999	3/2/0
HAR	linear	one_shard	5	1	0.68	3.39	0.999	0.999	4/1/0
HAR	mlp	spread	10	4	98.78	86.48	0.999	0.999	4/1/0
HAR	mlp	one_shard	5	1	21.70	108.83	0.999	0.999	2/3/0

TABLE S15
CLASS-LEVEL (MIRAI_UDP ON N-BAIOT) AND SAMPLE-LEVEL (25% OF THE LARGEST CLIENT) REQUESTS (5 SEEDS).

Data	Model	Kind	Method	Result	Rounds	Ckpt
N-BaIoT	linear	class	none	forget acc 1.000, retain acc 0.995	0	0
N-BaIoT	linear	class	retrain	forget acc 0.986, retain acc 0.994	60	0
N-BaIoT	linear	class	rollback	forget acc 0.986, retain acc 0.994	60	0
N-BaIoT	linear	sample	none	gap 1.04, acc 0.996	0	0
N-BaIoT	linear	sample	retrain	gap 0.96, acc 0.996	60	0
N-BaIoT	linear	sample	rollback	gap 0.96, acc 0.996	60	0
N-BaIoT	mlp	class	none	forget acc 1.000, retain acc 0.997	0	0
N-BaIoT	mlp	class	retrain	forget acc 1.000, retain acc 0.997	60	0
N-BaIoT	mlp	class	rollback	forget acc 1.000, retain acc 0.997	60	0
N-BaIoT	mlp	sample	none	gap 0.85, acc 0.997	0	0
N-BaIoT	mlp	sample	retrain	gap 1.11, acc 0.997	60	0
N-BaIoT	mlp	sample	rollback	gap 1.11, acc 0.997	60	0
HAR	mlp	sample	none	gap 1.16, acc 0.999	0	0
HAR	mlp	sample	retrain	gap 0.95, acc 0.999	60	0
HAR	mlp	sample	rollback	gap 0.95, acc 0.999	60	0

TABLE S16
ATTACKS ON THE AUDIT (MLP, 5 SEEDS): DETECTOR AUC, SHARE OF CANARIES FILTERED, UN-UNLEARNED MODEL FLAGGED HONESTLY, CERTIFIED AFTER FILTERING, CERTIFIED AFTER A SIGN-BALANCED SELECTIVE REVEAL UNDER THE ACCEPT-PARTIAL RULE AND UNDER THE STRICT RULE.

Data	Canary	AUC	Recall	Flagged	Filter cert.	Cherry (accept)	Cherry (strict)
HAR	gaussian	0.94	0.59	5/5	0/5	4/5	0/5
HAR	crossover	0.70	0.34	2/5	3/5	5/5	0/5
N-BaIoT	gaussian	0.92	0.48	5/5	2/5	5/5	0/5
N-BaIoT	crossover	0.71	0.28	1/5	4/5	5/5	1/5

TABLE S17
REWRITE-AND-RELINK ATTACK ON THE LEDGER (300 TRIALS PER INTERVAL, 3 WITNESSES). ED25519: 200/200 VALID, 200/200 TAMPERED CERTIFICATES REJECTED.

Anchor every	Chain only	With witnesses	Anchors	Receipt bytes
5	0.00	0.92	13	7800
10	0.00	0.89	7	4200
20	0.00	0.73	4	2400
60	0.00	0.02	2	1200

TABLE S18
CLOPPER–PEARSON 95% INTERVALS FOR HEADLINE RATES.

Quantity	k/n	%	95% CI
Exact retrains flagged, seeded order	4/280	1.4	0.4–3.6
Exact retrains flagged, random order	1/180	0.6	0.0–3.1
Early-stopped retrains flagged	1/25	4.0	0.1–20.4
Un-unlearned flagged, synthetic+HAR, seeded	160/160	100.0	97.7–100.0
Un-unlearned flagged, synthetic+HAR MLP, random	120/120	100.0	97.0–100.0
Un-unlearned flagged, N-BaIoT MLP, random	36/60	60.0	46.5–72.4
Selective reveal certified, accept-partial	19/20	95.0	75.1–99.9
Selective reveal certified, strict	1/20	5.0	0.1–24.9
Filter attack certified, Gaussian	2/10	20.0	2.5–55.6
Filter attack certified, crossover	7/10	70.0	34.8–93.3
retrain certified, Synthetic linear	5/5	100.0	47.8–100.0
retrain certified, HAR linear	3/5	60.0	14.7–94.7
retrain certified, HAR mlp	4/5	80.0	28.4–99.5
retrain certified, N-BaIoT linear	4/5	80.0	28.4–99.5
retrain certified, N-BaIoT mlp	5/5	100.0	47.8–100.0
retrain_early certified, Synthetic linear	5/5	100.0	47.8–100.0
retrain_early certified, HAR linear	4/5	80.0	28.4–99.5
retrain_early certified, HAR mlp	4/5	80.0	28.4–99.5
retrain_early certified, N-BaIoT linear	5/5	100.0	47.8–100.0
retrain_early certified, N-BaIoT mlp	5/5	100.0	47.8–100.0
icr_q1.0 certified, Synthetic linear	5/5	100.0	47.8–100.0
icr_q1.0 certified, HAR linear	4/5	80.0	28.4–99.5
icr_q1.0 certified, HAR mlp	4/5	80.0	28.4–99.5
icr_q1.0 certified, N-BaIoT linear	1/5	20.0	0.5–71.6
icr_q1.0 certified, N-BaIoT mlp	5/5	100.0	47.8–100.0
icr_q0.25 certified, Synthetic linear	4/5	80.0	28.4–99.5
icr_q0.25 certified, HAR linear	3/5	60.0	14.7–94.7
icr_q0.25 certified, HAR mlp	4/5	80.0	28.4–99.5
icr_q0.25 certified, N-BaIoT linear	1/5	20.0	0.5–71.6
icr_q0.25 certified, N-BaIoT mlp	5/5	100.0	47.8–100.0